Computer Networks

Unit 5: Network Protocols & Network Applications

Complete Exam-Ready Study Guide
Simple language • Full coverage • Diagrams • Practice questions

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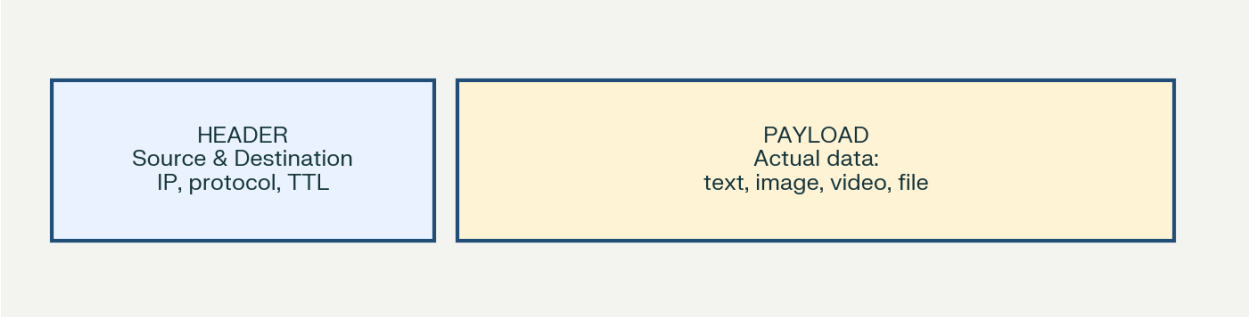
1. Introduction to Protocol and Packet

A **protocol** is a set of rules that controls communication between devices in a network. It decides what data is sent, how it is sent, and when it is sent.

A **packet** is a small unit of data that travels through a network. Large files are broken into many packets, sent separately, and reassembled at the destination.

Key Points

- Protocol = rules for network communication.
- Packet = header + payload.
- Header contains delivery information: source IP, destination IP, protocol, length, sequence number, TTL.
- Payload contains the actual data: text, image, video, or file.
- Packets may take different paths; lost or corrupted packets can be retransmitted.



Note: Think of a packet like a parcel: the header is the address label, and the payload is the item inside.

Important exam points: Define protocol; define packet; list the two parts of a packet; give examples of management, communication, and security protocols.

2. Internet Protocol (IP)

The **Internet Protocol (IP)** is the main protocol that delivers packets from a source device to a destination device using IP addresses.

Every device needs an IP address so data can reach the correct device. IP works at the **Network Layer (Layer 3)** of the OSI model and works with TCP or UDP.

Key Points

- IP uses source and destination IP addresses in the packet header.
- IP is **connectionless**: it does not create a dedicated connection before sending packets.
- IP provides best-effort delivery; reliability is handled by TCP when needed.
- Main versions: IPv4 and IPv6.
- IPv6 was introduced mainly because IPv4 addresses were becoming insufficient.

Important exam points: IP works at Layer 3; IP is connectionless; IPv4 is 32-bit and IPv6 is 128-bit.

3. IPv4 and its Header

IPv4 means Internet Protocol Version 4. It uses a 32-bit address written as four octets, for example 192.168.1.10. An IPv4 packet is called a **datagram** and has two parts: header and data.

The IPv4 header is at least 20 bytes and can grow to 60 bytes when optional fields are used.

Key Points

- **Version:** IP version; value is 4 for IPv4.
- **Header Length:** size of the header in 4-byte words.
- **Type of Service:** helps routers give priority or quality of service.
- **Total Length:** size of the complete packet including header and data.
- **Identification, Flags, Fragment Offset:** used for fragmentation and reassembly.
- **TTL:** maximum number of routers a packet can cross; each router reduces it by 1.
- **Protocol:** identifies the upper-layer protocol, such as TCP, UDP, or ICMP.
- **Header Checksum:** checks errors in the header.
- **Source and Destination Address:** sender and receiver IP addresses.
- **Options:** optional field for testing and debugging.

Note: If TTL becomes 0 before the packet reaches its destination, the packet is discarded. This prevents packets from looping forever.

Important exam points: Draw/explain IPv4 header; explain TTL; minimum header size is 20 bytes; fragmentation fields are Identification, Flags, and Fragment Offset.

4. IPv6 and its Header

IPv6 means Internet Protocol Version 6. It was developed mainly to solve the shortage of IPv4 addresses. It uses 128-bit addresses written in hexadecimal groups separated by colons.

The basic IPv6 header has a fixed size of 40 bytes, which makes processing simpler and faster for routers.

Key Points

- IPv6 address size: 128 bits.
- Basic header size: fixed 40 bytes.
- **Version:** always 6.
- **Traffic Class:** priority/type of traffic.
- **Flow Label:** identifies packets of the same flow, such as a video call.
- **Payload Length:** size of payload and extension headers.
- **Next Header:** identifies the next header or upper-layer protocol.
- **Hop Limit:** same purpose as TTL in IPv4.
- **Source/Destination Address:** 128-bit sender and receiver addresses.
- IPv6 has no header checksum.

Important exam points: IPv6 = 128-bit; header = 40 bytes; Hop Limit replaces TTL; no header checksum.

5. IPv4 vs IPv6

Use this table to answer comparison questions quickly.

Comparison / Table

Feature	IPv4	IPv6
Address size	32 bits	128 bits
Total addresses	2^{32}, about 4.3 billion	2^{12}■, extremely large
Format	Decimal with dots: 192.168.1.10	Hexadecimal with colons: 2001:db8::1
Header size	20–60 bytes, variable	Fixed 40 bytes
Header checksum	Present	Not present
Hop control	TTL	Hop Limit
Configuration	Manual or DHCP	Auto-configuration supported

Important exam points: The biggest difference is address size: 32 bits in IPv4 and 128 bits in IPv6.

6. IP Addresses and Classful Addressing

An IP address identifies a device on a network. An IPv4 address has 32 bits divided into four octets; each octet has 8 bits. It can be written in binary or dotted-decimal form.

Example: binary 10101100.00010000.11111110.00000001 equals decimal 172.16.254.1.

In classful addressing, IPv4 addresses are divided into Classes A, B, C, D, and E. Classes A, B, and C are used for normal one-to-one communication. Class D is for multicast, and Class E is reserved for experiments.

Key Points

- **Network ID:** identifies the network.
- **Host ID:** identifies a device inside the network.
- Class A: 0–127, 8 network bits, 24 host bits, large networks.
- Class B: 128–191, 16 network bits, 16 host bits, medium networks.
- Class C: 192–223, 24 network bits, 8 host bits, small networks.
- Class D: 224–239, multicast.
- Class E: 240–255, experimental/reserved.

Class A	0–127	Large networks	255.0.0.0 /8
Class B	128–191	Medium networks	255.255.0.0 /16
Class C	192–223	Small networks	255.255.255.0 /24
Class D	224–239	Multicast	—
Class E	240–255	Experimental	—

Note: To find the class quickly, check only the first number: 0–127 = A, 128–191 = B, 192–223 = C, 224–239 = D, 240–255 = E.

Important exam points: Class ranges; binary starting bits: A=0, B=10, C=110, D=1110, E=1111; identify class from an IP address.

7. Subnet Mask and Address Types

A **subnet mask** is a 32-bit number that separates an IP address into a network portion and a host portion. Network bits are written as 1 and host bits as 0.

Key Points

- Class A default mask: 255.0.0.0 = /8.
- Class B default mask: 255.255.0.0 = /16.
- Class C default mask: 255.255.255.0 = /24.
- **Network address:** first address; identifies the whole network.
- **Broadcast address:** last address; sends data to all hosts.
- **Host addresses:** addresses assigned to devices.
- Usable hosts = total addresses – 2, because network and broadcast addresses cannot be assigned to hosts.

Note: For 192.168.17.0/24: network address = 192.168.17.0, broadcast = 192.168.17.255, usable hosts = 192.168.17.1 to 192.168.17.254.

Important exam points: Default masks; identify network, broadcast, and usable host range; total addresses = 2^host bits.

8. Subnetting and CIDR

Subnetting means dividing a large network into smaller networks called subnets. It is done by borrowing bits from the host portion and using them as extra network bits.

CIDR means Classless Inter-Domain Routing. It writes an IP address with a prefix length, such as 192.168.1.0/26, instead of relying only on Classes A, B, and C.

Working / Steps

1. Find the default prefix of the address class: A=/8, B=/16, C=/24.
2. Subtract: borrowed bits = given prefix – default prefix.
3. Number of subnets = 2^borrowed bits.
4. Host bits = 32 – given prefix.
5. Total addresses per subnet = 2^host bits.
6. Usable hosts = 2^host bits – 2.
7. Write the subnet mask by putting 1s for network bits and 0s for host bits.

Key Points

- Example: 192.168.1.0/26 → borrowed bits = 26 – 24 = 2 → 4 subnets.
- Host bits = 32 – 26 = 6 → 64 total addresses → 62 usable hosts.
- Subnet mask = 255.255.255.192.
- Example: 192.168.1.0/27 → 3 borrowed bits → 8 subnets; 5 host bits → 32 addresses, 30 usable.
- Example: 139.10.10.1/20 → Class B default /16 → 4 borrowed bits → 16 subnets; 12 host bits → 4096 addresses, 4094 usable.

Note: The first address in each subnet is the network address and the last address is the broadcast address.

Important exam points: Calculate number of subnets, subnet mask, total addresses, usable hosts, and subnet ranges.

9. World Wide Web (WWW)

The **World Wide Web** is a collection of connected web pages and resources stored on web servers and accessed through the Internet. It includes text, images, audio, video, and documents.

Web pages are created using HTML and connected through hyperlinks. A URL gives the address of a web resource, and a website is a collection of related pages under one domain name.

Key Points

- WWW is a service that runs on the Internet.
- HTML structures web pages.
- Hyperlinks let users move from one page to another.
- HTTP/HTTPS are used to access web resources.
- Examples: www.google.com, www.facebook.com.

Note: Internet = the global network of devices. WWW = one service available on that network.

Important exam points: Define WWW; explain HTML, hyperlink, URL, and website; difference between Internet and WWW.

10. Web Hosting

Web hosting is a service that provides storage space and server resources to store website files and make them available on the Internet. Hosting companies keep servers running 24×7.

Because IP addresses are hard to remember, a domain name is connected to the server IP address using DNS.

Working / Steps

1. Register a unique domain name.
2. Choose a web hosting provider.
3. Configure the hosting server.
4. Upload website files.
5. Set up the database if required.
6. Configure DNS to connect the domain name to the server IP.
7. Publish and test the website.
8. Maintain it with updates, backups, and security checks.

Key Points

- **Shared hosting:** many websites share one server; cheap.
- **VPS hosting:** dedicated virtual resources on a shared server; more control.
- **Cloud hosting:** uses multiple connected servers; scalable and highly available.
- **Dedicated hosting:** one full physical server for one customer; maximum control and security.
- **Reseller hosting:** customer rents/sells hosting space to others.

Important exam points: Define web hosting; list workflow steps; explain hosting types with examples.

11. HTTP

HTTP means Hypertext Transfer Protocol. It is an application-layer protocol used for communication between a web client, usually a browser, and a web server.

The browser sends an HTTP request, and the server sends an HTTP response containing the requested web page or resource.

Working / Steps

1. The browser opens a TCP connection with the web server, normally on port 80.
2. The browser sends an HTTP request.
3. The server processes the request.
4. The server sends an HTTP response.
5. The browser displays the received web page.

Key Points

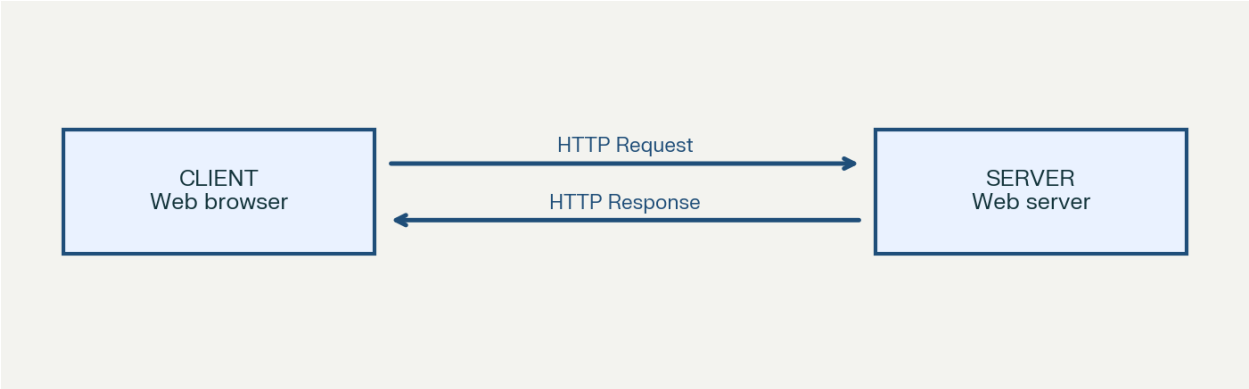
• HTTP uses TCP port 80; HTTPS is the secure version.

• HTTP is **stateless**: the server does not automatically remember previous requests.

• **Non-persistent HTTP:** one connection for one request-response, then it closes.

• **Persistent HTTP:** connection remains open for multiple requests and responses.

• Client examples: Chrome, mobile browser. Server example: Apache web server.



Important exam points: Explain HTTP request-response model; stateless protocol; difference between persistent and non-persistent HTTP.

12. Email, SMTP, POP3 and IMAP

Email is a digital method of sending and receiving messages over the Internet. It can include text, images, documents, and attachments.

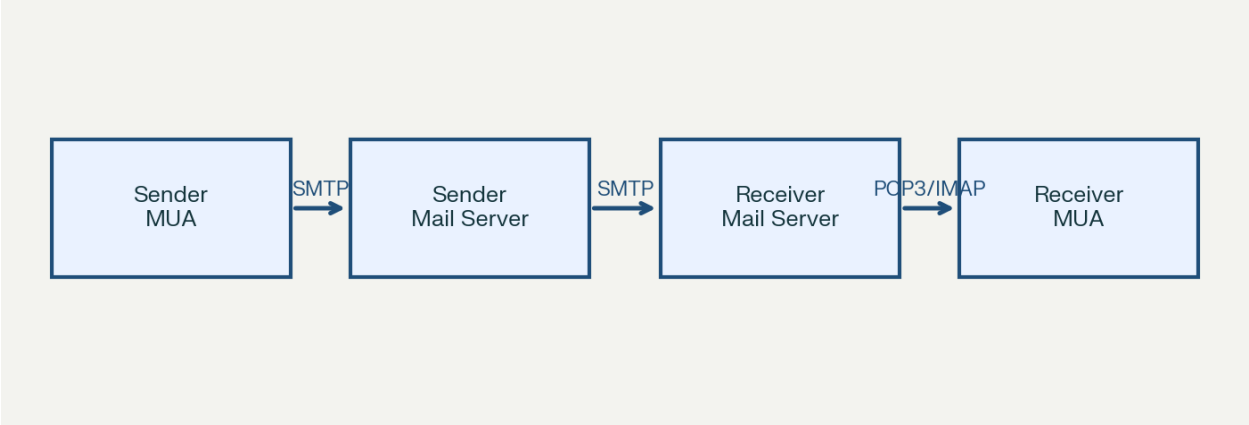
An email address has a username, @ symbol, and domain name, for example inform@gmail.com.

Working / Steps

1. Sender writes the email in an MUA.
2. The MUA sends it to the sender's mail server; it is placed in a queue.
3. SMTP opens a TCP connection with the receiver's mail server.
4. SMTP transfers the message.
5. The receiver's mail server stores the message in the mailbox.
6. The receiver reads it using an MUA through POP3 or IMAP.

Key Points

- **Header:** sender, receiver, subject, timestamp.
- **Body:** main message content.
- **Attachment:** attached file.
- **MUA:** Mail User Agent — Gmail, Outlook, Apple Mail; used to compose/read email.
- **MTA:** Message Transfer Agent — routes mail between mail servers.
- **SMTP** sends email using TCP port 25.
- **POP3** downloads email to the device.
- **IMAP** keeps and manages email on the server.



Note: If the receiver's mail server is down, SMTP keeps the message in the sender's mail server queue and tries again later.

Important exam points: Email components; MUA vs MTA; SMTP vs POP3 vs IMAP; SMTP port 25; explain email delivery with a diagram.

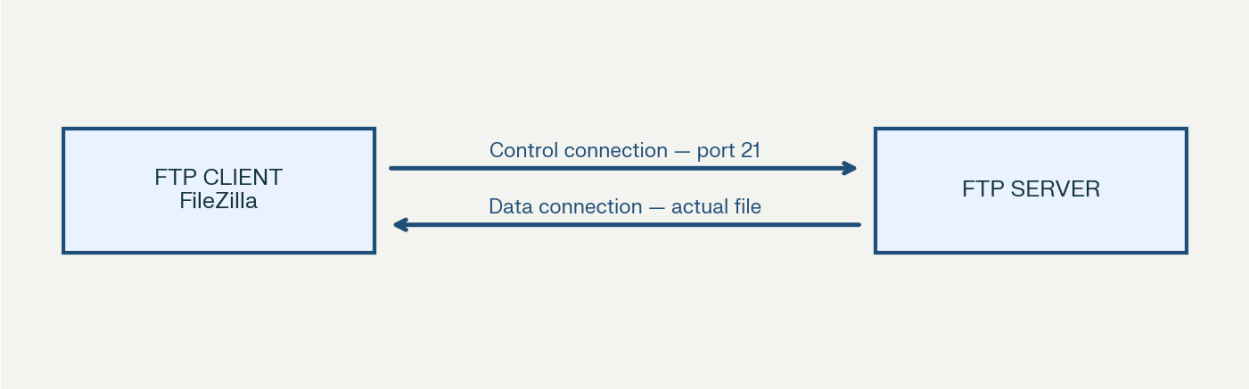
13. FTP

FTP means File Transfer Protocol. It transfers files between a client and a server over a TCP/IP network. FTP follows a client-server architecture and uses TCP for reliable transfer.

FileZilla is a common FTP client used to upload and download files.

Key Points

- FTP uses two separate TCP connections.
- **Control connection:** port 21; sends commands such as login, password, directory change, upload, and download.
- **Data connection:** transfers the actual file or data.
- FTP server maintains state, such as the current directory and authentication information.
- FTP is useful for uploading website files to a hosting server.



Important exam points: FTP uses two connections; control connection uses port 21; FTP is connection-oriented because it uses TCP.

14. DNS

DNS means Domain Name System. It converts a human-friendly domain name, such as www.google.com, into an IP address that computers use to communicate.

DNS is an application-layer protocol. If the answer is not in the resolver cache, DNS queries other servers until it finds the correct IP address.

Working / Steps

1. User enters a domain name in the browser.
2. The DNS resolver checks its cache.
3. If not found, it contacts a root server.
4. The root server directs it to the correct TLD server, such as .com.
5. The TLD server directs it to the authoritative server.
6. The authoritative server returns the IP address.
7. The browser connects to that IP address.

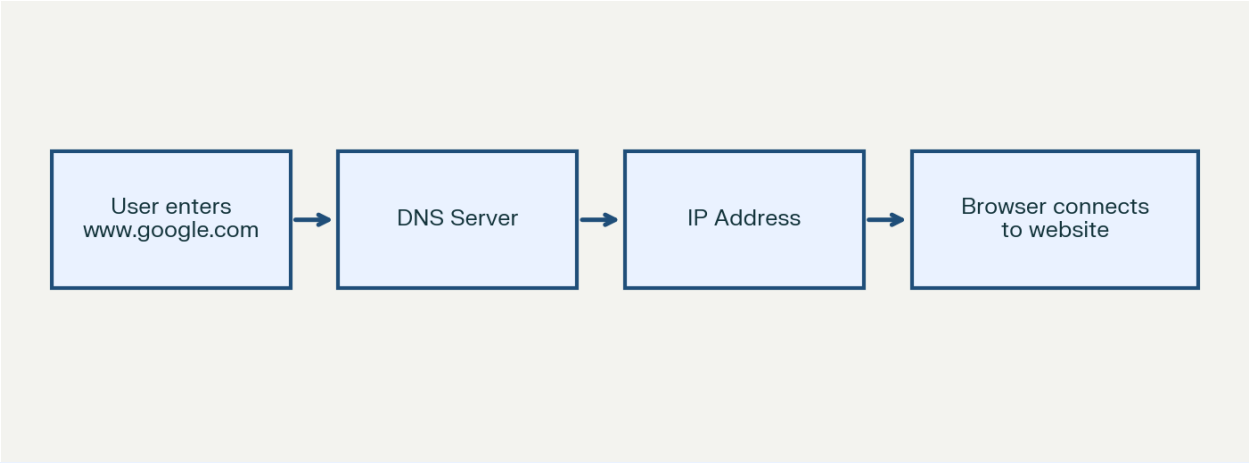
Key Points

• **Root servers:** first step; know the TLD servers; 13 root server systems worldwide.

• **TLD servers:** handle top-level domains such as .com, .org, .edu, .in.

• **Authoritative servers:** store the final hostname-to-IP mapping for an organization.

• DNS makes the Internet easy to use because people remember names better than IP addresses.



Important exam points: Define DNS; explain DNS hierarchy; explain DNS resolution steps; DNS is an application-layer protocol.

15. Network Applications

Network applications are programs that use a network to provide services to users.

Key Points

- Email — sending and receiving messages.
- Web — browsing websites.
- Remote login — accessing another computer.
- P2P file sharing — sharing files directly between users.
- Multi-user network games.
- Streaming stored video, such as YouTube.
- Voice over IP, such as internet calling.
- Real-time video conferencing.
- Social networking.

Important exam points: Give examples of network applications and the service each provides.

Short Answer Questions

Q. Define protocol.

Answer: A protocol is a set of rules that governs data communication between devices. It defines what, how, and when data is communicated.

Q. What is a packet?

Answer: A packet is a small unit of data sent over a network. It consists of a header, which contains delivery information, and a payload, which contains the actual data.

Q. Why was IPv6 developed?

Answer: IPv6 was developed mainly to overcome the shortage of IPv4 addresses. It provides a 128-bit address space.

Q. What is TTL?

Answer: TTL means Time To Live. It limits the number of routers a packet can pass through. Each router decreases TTL by 1; when it reaches 0, the packet is discarded.

Q. What is a subnet mask?

Answer: A subnet mask is a 32-bit number that separates an IP address into network bits and host bits. Network bits are 1 and host bits are 0.

Q. What is CIDR?

Answer: CIDR means Classless Inter-Domain Routing. It represents IP addresses using a prefix length, such as 192.168.1.0/27, instead of fixed class boundaries.

Q. Why is HTTP called stateless?

Answer: HTTP is stateless because the server does not automatically remember information about previous requests from the same client.

Q. What is DNS?

Answer: DNS is the Domain Name System. It converts human-readable domain names into IP addresses used by computers.

Q. What is the difference between SMTP and POP3?

Answer: SMTP is used to send email, while POP3 is used to receive/download email from a mail server.

Q. Why does FTP use two connections?

Answer: FTP uses a control connection for commands and authentication, and a separate data connection to transfer the actual files.

Long Answer Questions

Q. Explain IPv4 header format.

Answer: An IPv4 datagram has a header and data. The header is at least 20 bytes and includes Version, Header Length, Type of Service, Total Length, Identification, Flags, Fragment Offset, TTL, Protocol, Header Checksum, Source Address, Destination Address, and optional fields. Version identifies IPv4; TTL limits hops; Protocol identifies TCP/UDP/ICMP; checksum detects header errors; source and destination addresses identify sender and receiver.

Q. Compare IPv4 and IPv6.

Answer: IPv4 uses 32-bit addresses, about 4.3 billion total addresses, dotted-decimal notation, a variable 20–60 byte header, and a header checksum. IPv6 uses 128-bit addresses, hexadecimal colon notation, a fixed 40-byte header, Hop Limit instead of TTL, and no header checksum. IPv6 was created mainly to solve IPv4 address shortage.

Q. Explain subnetting with an example.

Answer: Subnetting divides a large network into smaller subnets by borrowing host bits. For 192.168.1.0/26, the Class C default prefix is /24, so borrowed bits = 2. Number of subnets = $2^2 = 4$. Host bits = 6, so each subnet has 2■ = 64 addresses and 62 usable hosts. The subnet mask is 255.255.255.192.

Q. Explain the working of HTTP.

Answer: A browser opens a TCP connection with a web server, usually on port 80. It sends an HTTP request; the server processes it and returns an HTTP response. The browser displays the response. HTTP is stateless. In non-persistent HTTP, the connection closes after one response; in persistent HTTP, it remains open for multiple requests.

Q. Explain email working with SMTP.

Answer: The sender writes an email using an MUA. The MUA sends it to the sender's mail server, where it is queued. SMTP opens a TCP connection with the receiver's mail server and transfers the message. The receiver's mail server places it in the mailbox. The receiver uses an MUA with POP3 or IMAP to read the message.

Q. Explain DNS name resolution.

Answer: The user enters a domain name. The DNS resolver checks its cache. If needed, it contacts a root server, which points to the TLD server. The TLD server points to the authoritative server for the domain. The authoritative server returns the IP address, and the browser connects to it.

MCQs with Answers

1. Which layer does IP work at?

a) Application layer

b) Transport layer

c) Network layer

d) Data link layer

Correct answer: c) Network layer — IP is a Layer 3, Network Layer protocol.

2. The payload of a packet contains:

a) Source IP address

b) Destination IP address

c) Actual data

d) TTL value

Correct answer: c) Actual data — The payload carries the actual data; the header carries control information.

3. IPv4 address size is:

a) 32 bits

b) 64 bits

c) 128 bits

d) 256 bits

Correct answer: a) 32 bits — IPv4 uses 32-bit addresses.

4. IPv6 header size is:

a) 20 bytes

b) 40 bytes

c) 60 bytes

d) 80 bytes

Correct answer: b) 40 bytes — The basic IPv6 header is fixed at 40 bytes.

5. Class B IP addresses range from:

a) 0–127

b) 128–191

c) 192–223

d) 224–239

Correct answer: b) 128–191 — Class B starts at 128 and ends at 191.

6. Default subnet mask of Class C is:

a) 255.0.0.0

b) 255.255.0.0

c) 255.255.255.0

d) 255.255.255.255

Correct answer: c) 255.255.255.0 — Class C uses /24, written as 255.255.255.0.

7. Number of subnets for /27 from a Class C /24 network:

a) 2

b) 4

c) 8

d) 16

Correct answer: c) 8 — Borrowed bits = 3, so subnets = $2^3 = 8$.

8. SMTP is mainly used for:

a) Receiving email

b) Sending email

c) Transferring files

d) Resolving domain names

Correct answer: b) Sending email — SMTP sends email messages.

9. FTP control connection uses TCP port:

a) 20

b) 21

c) 25

d) 80

Correct answer: b) 21 — FTP control connection uses port 21.

10. DNS converts:

a) IP address into MAC address

b) Domain name into IP address

c) IP address into domain name only

d) Files into packets

Correct answer: b) Domain name into IP address — DNS resolves domain names to IP addresses.

Final Revision Section

Concept	Remember
Protocol	Rules for communication
Packet	Header + payload
IP	Layer 3; connectionless; delivers packets
IPv4	32-bit; header 20–60 bytes; TTL; checksum
IPv6	128-bit; header 40 bytes; Hop Limit; no checksum
Class A/B/C	0–127 / 128–191 / 192–223
Class D/E	224–239 multicast / 240–255 reserved
Default masks	A /8, B /16, C /24
Subnetting	Borrow host bits; subnets = 2^N; addresses = 2^H; usable = 2^H – 2
HTTP	Web request/response; port 80; stateless
SMTP	Send email; TCP port 25
POP3/IMAP	Receive/manage email
FTP	File transfer; control port 21; uses TCP
DNS	Domain name → IP address; application layer

Last-Minute Checklist

- I can define protocol, packet, header, and payload.
- I can explain IP and state its OSI layer.
- I can draw/explain IPv4 and IPv6 headers.
- I can compare IPv4 and IPv6.
- I can identify IP class from the first octet.
- I can write default subnet masks.
- I can calculate subnets, total addresses, usable hosts, and ranges.
- I can explain WWW, web hosting, and hosting types.
- I can explain HTTP request-response and persistent/non-persistent connections.
- I can explain email flow and SMTP, POP3, IMAP.
- I can explain FTP control and data connections.
- I can explain DNS hierarchy and resolution.